

Weekly Report

Week no 02

Group no 03

Date : 12/7/2026

Project Repository: https://github.com/M000nwalker/Beyond_Horizon_1/tree/main/FEA%20Simulation

1 Simulation Setup (Comsol Multiphysics)

We modeled the objective lens structure as an achromatic doublet designed for refractive astronomical telescope applications. We collected material information from the manufacturer. We carried out computational ray tracing simulations to evaluate geometric optics performance, focus accuracy, and aberration behaviors across the visible spectrum under the following parametric configurations:

- **Ray Initiation Conditions:** We released a uniform parallel beam array along the positive z -axis from an entrance plane positioned exactly 100 mm below our local coordinate reference plane ($z = -100$ mm).
- **Spectral Distribution:** For our polychromatic analysis, we released rays of wavelengths spanning from $\lambda = 400$ nm to 700 nm, uniformly sampled at discrete increments of $\Delta\lambda = 50$ nm.

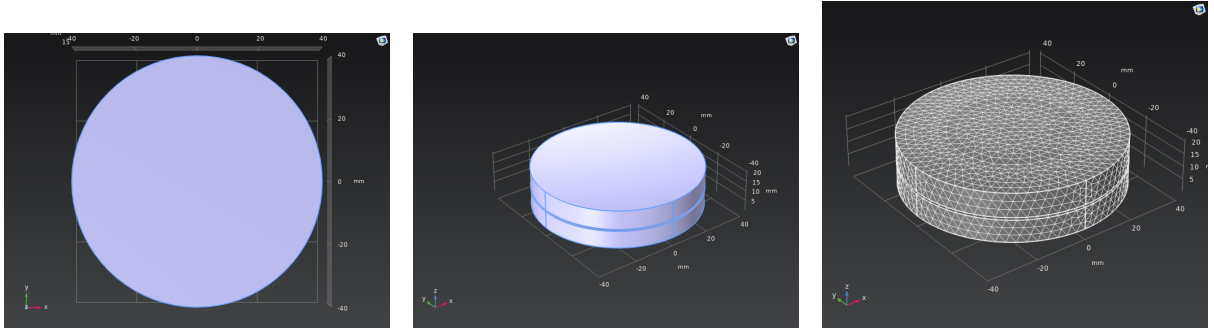


Figure 1: Top View, Isometric View and Meshing Of the Objective Lens

2 Simulation Findings & Analysis

Our numerical solution successfully converged, providing us with high-fidelity spatial data regarding focal point constraints, polychromatic ray paths, and multi-wavelength spot patterns.

We determined that an RMS spot radius of $68.1 \mu\text{m}$ across the complete visible bandwidth (400 nm – 700 nm) represents an excellent optical correction factor for an $f/11.25$ doublet, ensuring sharp point-source reconstructions highly suitable for high-resolution imaging and visual astronomy.

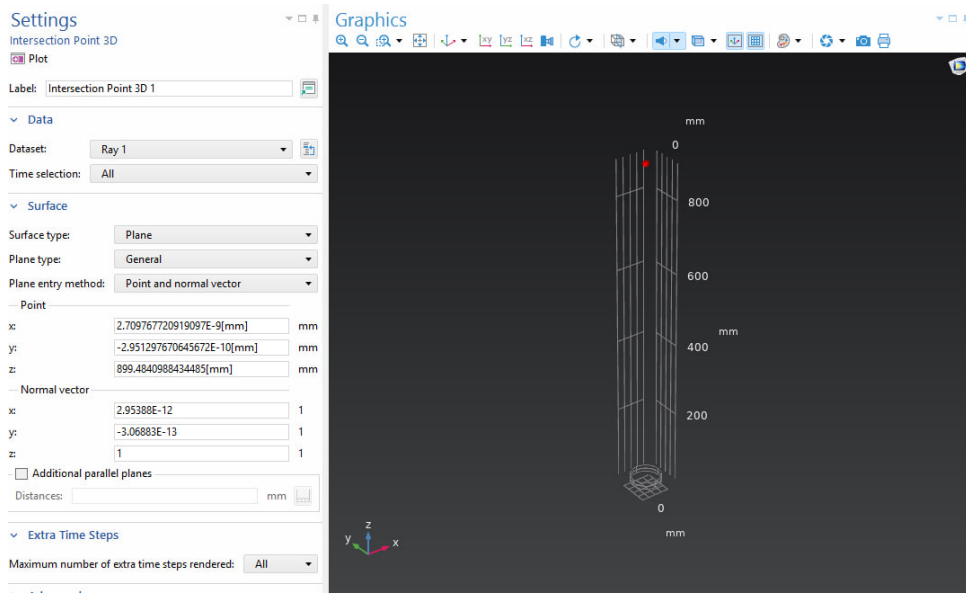


Figure 2: Focal Plane Dataset

In the above diagram, this shows our numerical intersection settings and coordinate measurement module, which verified a geometric paraxial focal convergence plane localized at $z = 899.484 \text{ mm}$, with negligible transverse beam walk ($x \approx 2.71 \times 10^{-9} \text{ mm}$, $y \approx -2.95 \times 10^{-10} \text{ mm}$).

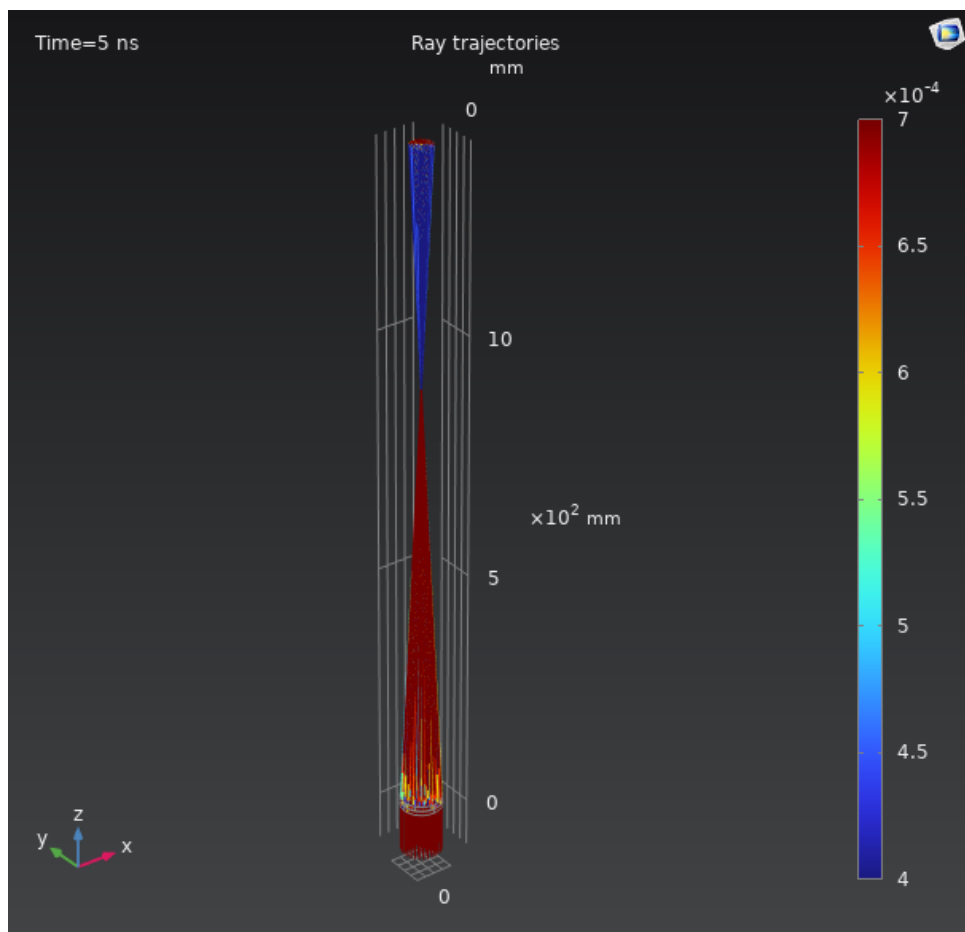


Figure 3: Ray Tracing

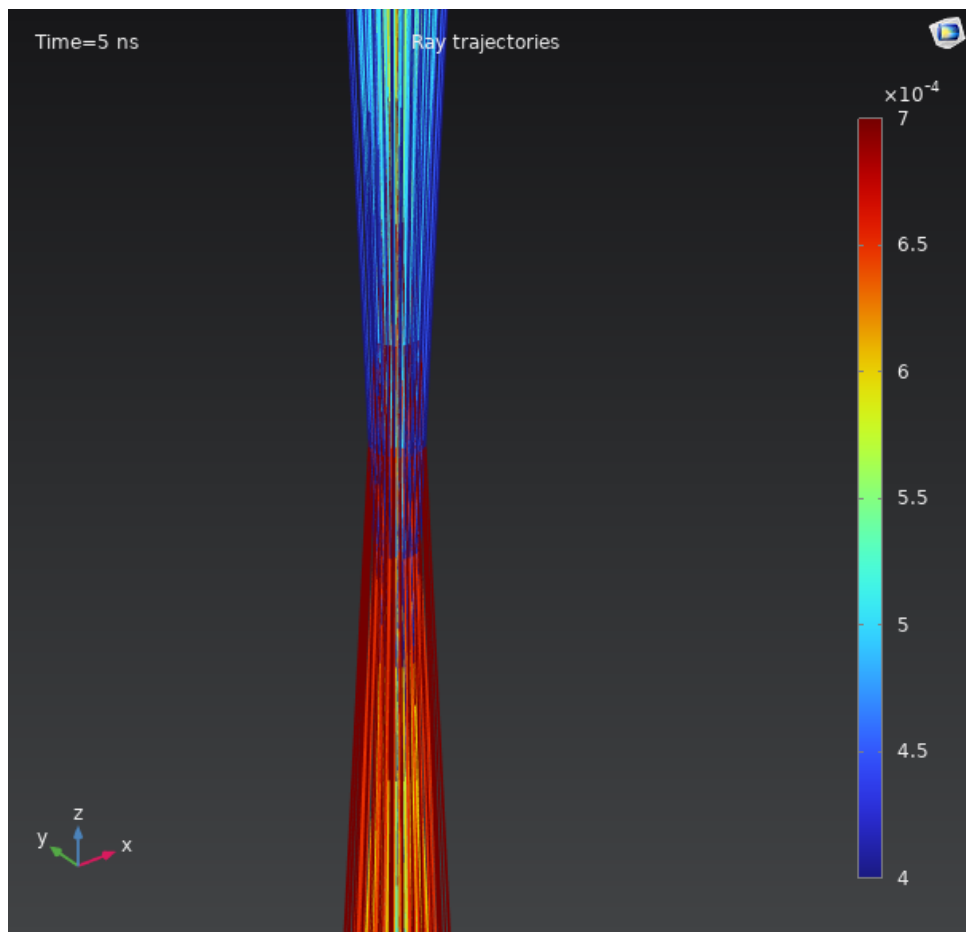


Figure 4: Ray Tracing (Highly Zoomed in)

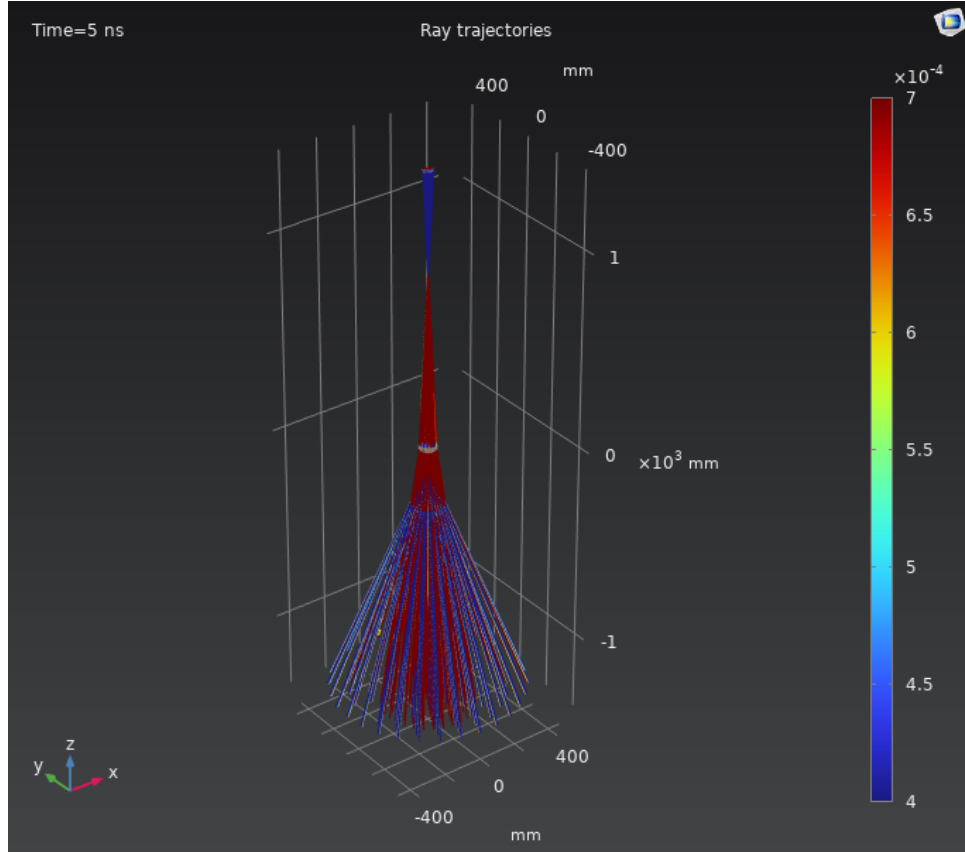


Figure 5: Ray Tracing Without Anti Reflective Coating

In the above diagram, this documents our comparative test configuration lacking thin-film coatings, where we observed prominent wide-angle secondary ghost reflections scattering backwards and outwards from the internal glass-to-glass and glass-to-air interfaces. We note that this stands in sharp contrast to the clean paths we saw in our primary AR-coated ray traces.

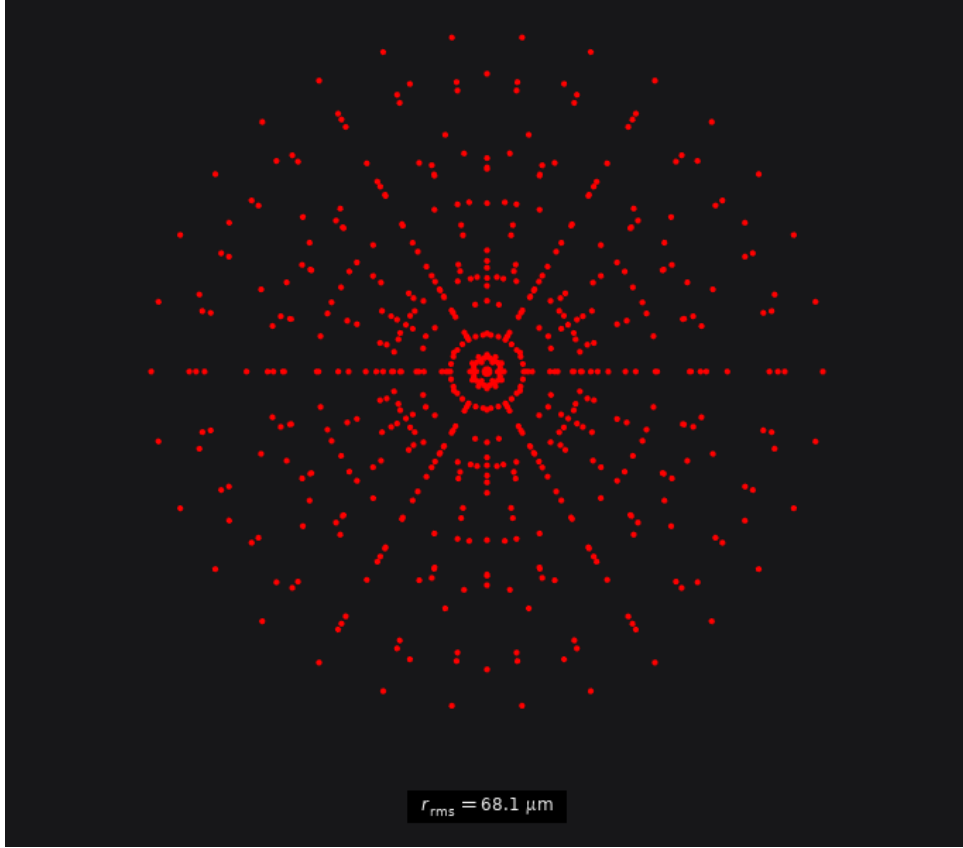


Figure 6: Spot Diagram

In the above diagram, this illustrates the final geometric intersection pattern of the sampled visible rays distributed on our optimal target focus plane. We found that the tightly packed symmetric cluster successfully maintains a root-mean-square spot radius of $r_{\text{rms}} = 68.1 \mu\text{m}$.